FR-07 – Step 3: Select test cases (best representative of each class)

Input: the classes in `FR-07_2_equivalence_classes.md`. Method: lecture S04, step 3: - Choose **at least one** test case from each equivalence class. - **Valid** classes: each test case covers as many valid classes as possible. - **Invalid** classes: each test case covers **one and only one** invalid class. All its other inputs come from valid classes.

Columns follow the course template `Test cases_Template_En.xlsx`. Actual result, status, tester and date are filled in at step 5 (execution).

Test data

Products as listed on the EShop home page after seeding (visible in the UI):

Name	Product	Unit price
A	iPhone 15 Pro Max	30.000.000 ₫
B	Samsung Galaxy S24 Ultra	28.000.000 ₫
C	MacBook Pro M3	45.000.000 ₫

Product IDs (for the API) are read from the product detail URL `/product/<id>`; below they are assumed to be A = 1, B = 2, C = 3. Money may be shown with `.` or `,` as the thousand separator; either satisfies FR-21 as long as it is consistent.

Common setup

ID	Setup
P1	Backend (`:3000`) and frontend-web (`:5173`) running; database freshly seeded (`node database.js`). Cart is empty.
P2	Logged in as `test@eshop.com` / `Test1234!`.
P3	Guest: not logged in (log out, or use a private window).
P4	**M is known.** M is the largest quantity the detail page accepts, found by FR-07_TC_24. Test cases marked P4 run after TC_24.
API	Postman or curl. Get a token with `POST http://localhost:3000/api/login` (`test@eshop.com` / `Test1234!`). `POST /api/cart` adds an item; `GET /api/cart` shows the server-side cart. Header: `Authorization: Bearer <token>`.
Cleanup	After each test case, empty the cart (delete every row) so the next one starts from an empty cart.

Valid test cases

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-07_TC_01	Add a product with quantity 2 and view the cart (logged in)	EC_01, 03, 05, 11, 15, 17, 29, 32; outputs EC_63, 64, 65, 66, 67, 68, 74, 76, 77, 84, 85, 86	P1, P2	1. Open A's detail page. 2. Quantity `2`, click "Add to cart". 3. Open the Cart page.	2. Toast or badge update shown; badge = 2. 3. One row. Columns exactly Sản phẩm, Đơn giá, Số lượng, Thành tiền, Thao tác. Unit price 30.000.000 ₫; quantity 2; line total 60.000.000 ₫. Total labelled "Tổng cộng" = 60.000.000 ₫. Button "Tiếp tục mua sắm". Delete button red. Breadcrumb present. Exactly one `<h1>`.
FR-07_TC_02	Add the same product again from the same page	EC_18; outputs EC_65, 69, 77	P1, P2; cart has A × 2 (from the detail page)	1. Open A's detail page, quantity `1`, "Add to cart". 2. Open the Cart.	Still **one** row for A, quantity 3, line total 90.000.000 ₫; badge = 3.
FR-07_TC_03	Guest adds two products from the home page	EC_16, 30, 31; output EC_67	P1, P3	1. On the home page click "Add to cart" for A, then for B. 2. Open the Cart.	Guest is allowed (D1). Two rows, quantity 1 each. Total 58.000.000 ₫; badge = 2.
FR-07_TC_04	Increase and decrease quantity with + and −	EC_20, 22; outputs EC_65, 66, 67, 77	P1, P2; cart has A × 2	1. Click "+". 2. Click "−".	1. Quantity 3; line total and total 90.000.000 ₫; badge 3. 2. Quantity 2; totals 60.000.000 ₫; badge 2.
FR-07_TC_05	Delete a row and confirm	EC_24, 25; outputs EC_70, 71	P1, P2; cart has A × 1 and B × 1	1. Click Delete on A. 2. Confirm.	1. Confirmation dialog shown. 2. A removed; B remains; total 28.000.000 ₫; badge 1.
FR-07_TC_06	Delete a row and cancel	EC_26; output EC_72	P1, P2; cart has A × 1	1. Click Delete on A. 2. Cancel.	A still in the cart, unchanged.
FR-07_TC_07	Delete the last row	EC_28; outputs EC_75, 86	P1, P2; cart has A × 1	1. Delete A and confirm.	Empty state: illustration/icon and a clear message; exactly one `<h1>`; badge 0 or hidden.
FR-07_TC_08	"Continue shopping", then come back	EC_27, 33; outputs EC_73, 79	P1, P2; cart has A × 2	1. Click "Tiếp tục mua sắm". 2. Open the Cart from the navbar.	1. Home page shown. 2. A × 2 still in the cart.
FR-07_TC_09	Type a quantity in the cart	EC_55	P1, P2; cart has A × 2	Type `5` in A's quantity field, confirm (Enter / leave the field).	Quantity 5; line total and total 150.000.000 ₫; badge 5.

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-07_TC_10	Type a quantity with leading zeros in the cart	EC_62	P1, P2; cart has A × 2	Type `007` in A's quantity field, confirm.	Quantity 7 (tester decision); totals 210.000.000 ₫.
FR-07_TC_11	Detail page quantity with leading zeros	EC_08	P1, P2	On A's detail page enter `007`, "Add to cart".	A × 7 added (tester decision).
FR-07_TC_12	Full page reload, logged in	EC_34 (logged in); output EC_79	P1, P2; cart has A × 2	Press F5 on the Cart page.	A × 2 still in the cart (tester decision D5).
FR-07_TC_13	Full page reload, guest	EC_34 (guest); output EC_80	P1, P3; cart has A × 1	Press F5 on the Cart page.	Cart kept or emptied; **both acceptable** for a guest (D5). Record which.
FR-07_TC_14	Log out and log in again	EC_35; output EC_79	P1, P2; cart has A × 2	1. Log out. 2. Log in as `test@eshop.com`. 3. Open the Cart.	A × 2 still in the cart (tester decision D5).
FR-07_TC_15	Add to cart through the API with correct data	EC_36, 40, 43, 49, 52	P1, API	`POST /api/cart` `{"id":1,"name":"iPhone 15 Pro Max","price":30000000,"quantity":2}`, then `GET /api/cart`.	HTTP 200; the cart contains A × 2 at 30.000.000 ₫.
FR-07_TC_16	Same product from the home page, then the detail page	EC_19 (decided at execution)	P1, P2	1. Home page: "Add to cart" for A. 2. A's detail page: quantity `2`, "Add to cart". 3. Open the Cart.	**Decided at execution (D8).** Expected if the cart is kept between pages: one row, A × 3. Record what happens.

Invalid test cases – quantity on the detail page

Precondition for all: P1, P2, on A's detail page, empty cart. Each test enters one quantity and clicks "Add to cart". The field uses `type="number"` on many browsers; if the browser blocks the value from being typed at all, record that: it also counts as rejected.

TC ID	Test case name	Class covered	Quantity entered	Expected result
FR-07_TC_17	Empty quantity	EC_04; output EC_78	`""`	Invalid-quantity error; cart and badge unchanged.

TC ID	Test case name	Class covered	Quantity entered	Expected result
FR-07_TC_18	Decimal quantity	EC_06	`1.5`	Invalid-quantity error; cart unchanged.
FR-07_TC_19	Non-numeric quantity	EC_07	`abc`	Rejected (error, or cannot be typed); cart unchanged.
FR-07_TC_20	Quantity with plus sign	EC_09	`+3`	Invalid-quantity error (tester decision); cart unchanged.
FR-07_TC_21	Quantity in scientific notation	EC_10	`1e2`	Invalid-quantity error (tester decision); cart unchanged.
FR-07_TC_22	Quantity 0	EC_12	`0`	Invalid-quantity error; cart unchanged.
FR-07_TC_23	Negative quantity	EC_13	`-1`	Invalid-quantity error; cart unchanged.
FR-07_TC_24	Quantity above the maximum (find M)	EC_14	`99999999999` ; if accepted, empty the cart and try ×10 (`9999999999 90` , …) until rejected or `1e21` is reached	A maximum exists: the first rejected value shows an invalid-quantity error. **Record M** = the largest value accepted. If nothing is rejected up to `1e21` , record "no limit" as the finding (D2).

Invalid test cases – + and − buttons

TC ID	Test case name	Class covered	Precondition	Test steps	Expected result
FR-07_TC_25	"+" at the maximum	EC_21	P1, P2, P4; cart has A × M	Click "+".	Quantity stays M; error shown or "+" disabled.
FR-07_TC_26	"−" at quantity 1	EC_23; output EC_87	P1, P2; cart has A × 1	Look at "−", then try to click it.	"−" is disabled; the row stays with quantity 1 (tester decision D3).

Invalid test cases – quantity typed in the cart

Precondition for all: P1, P2, cart has A × 2. Each test types one value in A's quantity field and confirms.

TC ID	Test case name	Class covered	Value typed	Expected result
FR-07_TC_27	Empty	EC_56	`""`	Invalid-quantity error; quantity goes back to 2; totals unchanged.
FR-07_TC_28	Zero	EC_57	`0`	Invalid-quantity error; quantity stays 2 (row is not deleted).
FR-07_TC_29	Negative	EC_58	`-1`	Invalid-quantity error; quantity stays 2.
FR-07_TC_30	Above the maximum	EC_59	M + 1 (needs P4)	Invalid-quantity error; quantity stays 2.
FR-07_TC_31	Decimal	EC_60	`2.5`	Invalid-quantity error; quantity stays 2.
FR-07_TC_32	Non-numeric	EC_61	`abc`	Rejected; quantity stays 2.
FR-07_TC_33	Plus sign	EC_88	`+3`	Invalid-quantity error (tester decision); quantity stays 2.

Invalid test case – product removed from the catalog

TC ID	Test case name	Class covered	Precondition	Test steps	Expected result
FR-07_TC_34	Product in the cart is deleted by admin	EC_02	P1, P2; cart has C × 1; admin web app running	1. In the admin app, delete product C. 2. As the user, open (or reload) the Cart.	C's row is removed or marked unavailable and cannot be checked out (tester decision).

Invalid test cases – cart API

Precondition for all: P1, API (valid token unless stated). Each request is the valid body from TC_15, `{"id":1,"name":"iPhone 15 Pro Max","price":30000000,"quantity":2}`, with **one** field changed. After each request, check `GET /api/cart`.

TC ID	Test case name	Class covered	Change from the valid request	Expected result
FR-07_TC_35	Lower price	EC_37; output EC_82	`"price":1`	Cart shows A at the catalog price 30.000.000 ₫, not 1 (D7).
FR-07_TC_36	Negative price	EC_38	`"price":-30000000`	Catalog price used; no negative total.
FR-07_TC_37	Non-numeric price	EC_39	`"price":"abc"`	Catalog price used, or request rejected; never `NaN`.
FR-07_TC_38	Different name	EC_41	`"name":"Free iPhone"`	Cart shows the catalog name "iPhone 15 Pro Max".
FR-07_TC_39	Name with HTML	EC_42	`"name":"<img src=x onerror=alert(1)>"`	Catalog name shown; in the cart page no image tag is rendered and no script runs (SEC-04).
FR-07_TC_40	Quantity 0	EC_44; output EC_83	`"quantity":0`	Request rejected (HTTP 4xx); cart unchanged.
FR-07_TC_41	Negative quantity	EC_45	`"quantity":-5`	Rejected; cart unchanged.
FR-07_TC_42	Quantity above the maximum	EC_46	`"quantity":` M + 1 (needs P4)	Rejected; cart unchanged.
FR-07_TC_43	Decimal quantity	EC_47	`"quantity":1.5`	Rejected; cart unchanged.
FR-07_TC_44	Non-numeric quantity	EC_48	`"quantity":"abc"`	Rejected; cart unchanged.
FR-07_TC_45	Non-existent product ID	EC_50	`"id":99999`	Rejected; cart unchanged.
FR-07_TC_46	Non-numeric product ID	EC_51	`"id":"abc"`	Rejected; cart unchanged.

TC ID	Test case name	Class covered	Change from the valid request	Expected result
FR-07_TC_47	No token	EC_53; output EC_81	Remove the `Authorization` header	HTTP 401/403; cart unchanged.
FR-07_TC_48	Tampered token	EC_54	Change one character in the token's signature part	HTTP 401/403; cart unchanged.

Coverage check

Every class from step 2 is covered by at least one test case, and each invalid test case targets exactly one invalid input class.

Class	Test case		Class	Test case		Class	Test case
EC_01	TC_01		EC_31	TC_03		EC_61	TC_32
EC_02	TC_34		EC_32	TC_01		EC_62	TC_10
EC_03	TC_01		EC_33	TC_08		EC_63	TC_01
EC_04	TC_17		EC_34	TC_12, TC_13		EC_64	TC_01
EC_05	TC_01		EC_35	TC_14		EC_65	TC_01
EC_06	TC_18		EC_36	TC_15		EC_66	TC_01
EC_07	TC_19		EC_37	TC_35		EC_67	TC_01
EC_08	TC_11		EC_38	TC_36		EC_68	TC_01
EC_09	TC_20		EC_39	TC_37		EC_69	TC_02
EC_10	TC_21		EC_40	TC_15		EC_70	TC_05
EC_11	TC_01		EC_41	TC_38		EC_71	TC_05
EC_12	TC_22		EC_42	TC_39		EC_72	TC_06
EC_13	TC_23		EC_43	TC_15		EC_73	TC_08
EC_14	TC_24		EC_44	TC_40		EC_74	TC_01
EC_15	TC_01		EC_45	TC_41		EC_75	TC_07
EC_16	TC_03		EC_46	TC_42		EC_76	TC_01
EC_17	TC_01		EC_47	TC_43		EC_77	TC_01
EC_18	TC_02		EC_48	TC_44		EC_78	TC_17
EC_19	TC_16		EC_49	TC_15		EC_79	TC_08
EC_20	TC_04		EC_50	TC_45		EC_80	TC_13
EC_21	TC_25		EC_51	TC_46		EC_81	TC_47
EC_22	TC_04		EC_52	TC_15		EC_82	TC_35
EC_23	TC_26		EC_53	TC_47		EC_83	TC_40
EC_24	TC_05		EC_54	TC_48		EC_84	TC_01
EC_25	TC_05		EC_55	TC_09		EC_85	TC_01
EC_26	TC_06		EC_56	TC_27		EC_86	TC_01
EC_27	TC_08		EC_57	TC_28		EC_87	TC_26

Class	Test case		Class	Test case		Class	Test case
EC_28	TC_07		EC_58	TC_29		EC_88	TC_33
EC_29	TC_01		EC_59	TC_30			
EC_30	TC_03		EC_60	TC_31			

Summary

	Count
Valid test cases	15
Test case with expected result decided at execution	1 (TC_16)
Invalid test cases	32
Total test cases	**48**
Classes covered	88 / 88

Notes for review

1. **TC_01 checks 12 outputs at once** (columns, labels, colours, breadcrumb, heading, prices). If it fails, list each failing output separately in the bug reports.
2. **TC_24 finds M.** TC_25, TC_30 and TC_42 depend on it. If no limit is found, those three are "Not executed" and "no upper limit" is the finding.
3. **The API test cases check the server-side cart** (`GET /api/cart`). Whether the cart page in the browser shows the same data is itself worth recording at execution.
4. **TC_34 needs the admin app** (FR-15 product delete). If deleting fails, mark TC_34 "Not executed" with the reason.
5. **Expected error wording** is not in the spec; "invalid-quantity error" means "a message with that meaning".
6. **Test data comes from what the home page shows after seeding.** If the products or prices differ on your machine, update the totals before executing.